

Dampers and Intro to Laplace

EGR 345

GVSU

Outline

- quick review
- clean-up vertical mass spring EOM
- what is a damper?
- intro to Laplace

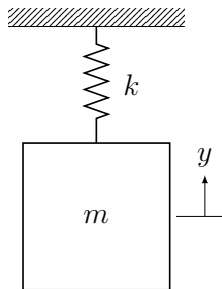
Quick Review

- what is this class about?

Quick Review

- what is this class about?
- what is a dynamic system?
- what is feedback control?
- why are we focusing on mass/spring/damper systems?

Vertical MSD Clean-Up



$$\Delta y = y - y_{\text{initial}}$$

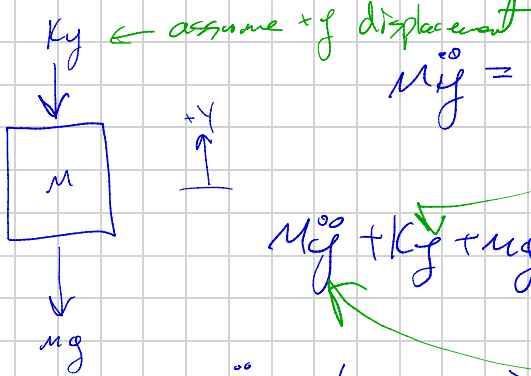
↓
0

$$f_{\text{spring}} = ky$$

- [assume y is measured from the undeformed length of the spring]
- define $\hat{y}$ as the distance from the point where gravity and the spring force cancel:

$$y = \hat{y} + y_{\text{static}}$$

← from gravity



$$m\ddot{y} = -k_y - mg$$

$$m\ddot{y} + k_y + mg = 0$$

$$m\ddot{\hat{y}} + k(\hat{y} + y_{\text{STATIC}}) + mg = 0$$

$$m\ddot{\hat{y}} + k\hat{y} + k y_{\text{STATIC}} + mg = 0$$

$$y = \hat{y} + y_{\text{STATIC}}$$

$$\dot{y} = \dot{\hat{y}}$$

$$\ddot{y} = \ddot{\hat{y}}$$

$$y_{\text{STATIC}} = \frac{-mg}{k}$$



if $\hat{y}$ is measured from static equilibrium:

$$m\ddot{\hat{y}} + k\hat{y} = 0$$

Equation of Motion (EOM)

LA1: given $y(0) = \#$, find $y(t)$

What is a damper?

- what did you learn from the video?
- why do suspensions include a damper?
 - what happens to mass/spring systems that do not have a damper?
- how do we find the force of the damper mathematically?
 - $f = kx$ for a spring
 - ~~$f = kx$~~ ? what a damper?

f

force is proportional to
velocity
 $f = b\dot{x}$

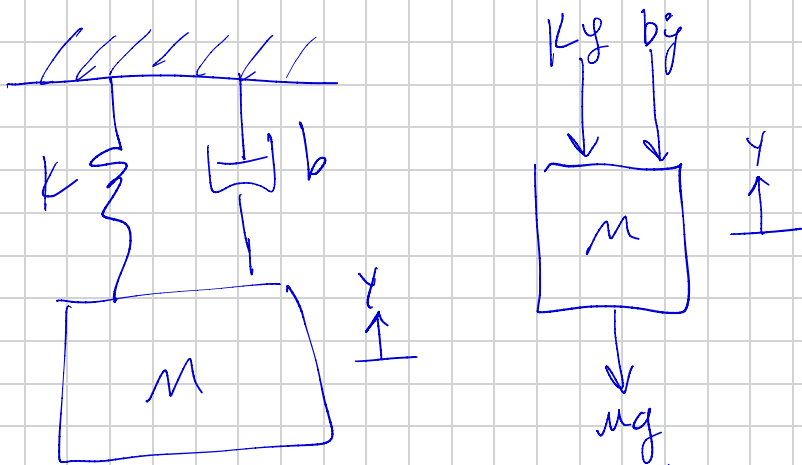
→ represent energy dissipation

Damper Experiment

- what do we notice about the response of the system without a damper?
- what is different once the damper is included?

↑
vibration dies out
much faster

↑
seems to vibrate
nearly forever



$$m\ddot{y} = -ky - by - mg$$

$$m\ddot{y} + by + ky + mg = 0$$

$$m\ddot{y} + b\dot{y} + ky + mg = 0$$

$$y = \hat{y} + y_{\text{STATIC}}$$

$$m\ddot{\hat{y}} + b\dot{\hat{y}} + k\hat{y} + \underbrace{ky_{\text{STATIC}} + mg}_{=0} = 0$$

$$y_{\text{STATIC}} = \frac{-mg}{k}$$

$$\boxed{m\ddot{\hat{y}} + b\dot{\hat{y}} + k\hat{y} = 0} \quad \text{EOM}$$

↑
damping term

Intro to Laplace

Background and History of Laplace and Fourier Transforms

- Laplace History:

- https://en.wikipedia.org/wiki/Laplace_transform#History

- Fourier History: *OTF Fourier: any periodic signal can be represented as a series of sin and cosine terms*

- https://en.wikipedia.org/wiki/Fourier_transform#History

- **Key Ideas:**

- dudes in the 1800's played around with integrals for funsies
 - the internet hadn't been invented yet
- some of their integrals turned out to be really helpful

Definition of the Laplace Transform

$$\mathcal{L}[f(t)] = \int_0^{\infty} f(t)e^{-st}dt$$

- unless you are an unbelievable mathematical genius, this shouldn't mean anything yet

How does the Laplace transform help us?

Key property of the Laplace transform:

- the Laplace transform turns differential equations into algebra

$$\frac{df}{dt} \rightarrow sF$$

(assuming zero initial conditions)

STC multiplying by s in the Laplace domain is the same as differentiating with respect to time

Foundational Laplace Transform

$$\mathcal{L}[f(t)] = F(s)$$

$$\mathcal{L}\left[\frac{df}{dt}\right] = sF(s) - f(0)$$

- let's derive this!
- math is fun

the Laplace transform of $\frac{df}{dt}$ is

s times the Laplace transform
of $f(t) - f(0)$

→ differentiation is replaced by multiplication by s

Goal: prove $\mathcal{L}\left[\frac{df}{dt}\right] = s f(s) - f(0)$

$$\mathcal{L}[f(t)] = \int_0^{\infty} f(t) e^{-st} dt$$

$$\mathcal{L}\left[\frac{df}{dt}\right] = \int_0^{\infty} \frac{df}{dt} \cdot e^{-st} dt$$

integration by parts: $\int u dv = u \cdot v - \int v du$

$$\int_0^{\infty} \frac{df}{dt} \cdot e^{-st} dt = \int_0^{\infty} u dv = u \cdot v - \int_0^{\infty} v du$$

$$\int \left(dv = \frac{df}{dt} dt \right) \quad u = e^{-st}$$

$$v = f$$

$$du = -s e^{-st} dt$$

$$\underbrace{\int_0^{\infty} \frac{df}{dt} e^{-st} dt}_{\int \left(\frac{df}{dt} \right)} = e^{-st} \cdot f \Big|_0^{\infty} - \int_0^{\infty} f (-s e^{-st}) dt$$

$$\mathcal{L}\left[\frac{df}{dt}\right] = \left. e^{-st} \cdot f \right|_0^{\infty} - \int_0^{\infty} f(t) (-s e^{-st}) dt$$

$$\mathcal{L}\left[\frac{df}{dt}\right] = \cancel{e^{-s\infty}} f(\infty) - \cancel{e^{-s\cdot 0}} \cdot f(0) + s \underbrace{\int_0^{\infty} f(t) e^{-st} dt}_{F(s)}$$

$$\therefore \mathcal{L}\left[\frac{df}{dt}\right] = s F(s) - f(0)$$

Laplace example LA 1

1 2 3 4

$\{K$

M

$$\mathcal{L} [m \ddot{y} + K y = 0]$$

$$\hat{y}(0) = y_0$$

$$\dot{\hat{y}}(0) = 0$$

$$\text{row 13: } \mathcal{L} [\ddot{y}] = s^2 \hat{Y}(s) - s \hat{y}(0) - \dot{\hat{y}}(0)$$

$$\mathcal{L} [\hat{y}] = \hat{Y}(s)$$

$$m(s^2 \hat{Y} - s y_0) + K \hat{Y} = 0$$

$$m(s^2 \hat{y} - sy_0) + k\hat{y} = 0$$

$$\hat{y} = ?$$

$$ms^2 \hat{y} - msy_0 + k\hat{y} = 0$$

$$\hat{y}(ms^2 + k) = msy_0$$

$$\hat{y} = \frac{msy_0}{ms^2 + k} \cdot \frac{1}{1/m}$$

$$\hat{y} = y_0 \frac{s}{s^2 + \frac{k}{m}}$$

$$\omega^2 = \frac{k}{m}$$

$$\omega = \sqrt{\frac{k}{m}}$$

$$\hat{y}(s) = y_0 \frac{s}{s^2 + \omega^2}$$

$$\hat{y}(s) = y_0 \frac{s}{s^2 + \omega^2}$$

inverse
Laplace $\mathcal{L}^{-1}$



row 7
row 10

$$\hat{y}(t) = y_0 \cos(\omega t)$$

FBD



EOM (diff. eq.)



Laplace



algebra



inverse Laplace (table) $\rightarrow \hat{y}(t) = \underline{\hspace{2cm}}$

Notation clarification:

$$\frac{dy}{dt} = \dot{y} \text{ (velocity)} \quad \frac{d^2y}{dt^2} = \ddot{y} \text{ (acceleration)}$$

$$y = \hat{y} + y_{\text{static}}$$

↑ measured from a different
Starting point (static equilibrium)

Learning Activity 1

- review some diff. eq.

Laplace Analysis for LA 1

- probably won't have time for this

Done